

AI-Enabled IoT Smart Energy Meter

Prototype Research Progress Report: IoT Monitoring and Predictive Energy Analytics

Current prototype status Hardware assembled and live readings demonstrated	Communication layer ESP32 Wi-Fi connection with Blynk dashboard
Measured variables Voltage, current, power and cumulative energy	Next research stage Calibrated logging, analysis and next-day prediction

This report documents the actual working prototype demonstrated on 18 September 2026. It separates verified prototype observations from planned research work and records the required calibration and data-collection steps.

Prepared for project review

1. Abstract and Project Objective

This project develops an ESP32-based smart energy meter that acquires electrical measurements and makes them available on a local LCD and a Blynk mobile dashboard. The completed prototype establishes the data-acquisition and IoT-monitoring foundation. The next phase will store calibrated time-series readings, analyse historical consumption, and predict the following day's electricity consumption.

Verified progress to date

The provided evidence shows a breadboard prototype using an ESP32, a 16x2 I2C LCD, ACS712 current sensing, a ZMPT101B voltage sensor, a connected load, and a Blynk mobile dashboard. The serial monitor shows repeatedly refreshed voltage, current, power and cumulative-energy values. The dashboard presents the same four readings remotely.

Objective	Current evidence / next step
Measure electrical parameters	Live voltage, current and power are shown on serial output, LCD and Blynk.
Retain accumulated reading	Firmware uses ESP32 EEPROM to retain the unit value after restart.
Remote visibility	Blynk virtual pins V0-V3 supply the mobile dashboard.
Predict next-day consumption	Planned after calibrated time-stamped data collection is available.

Research question

Can a low-cost ESP32 energy meter provide sufficiently reliable time-series data for an IoT dashboard and short-term, next-day household energy-consumption forecast?

2. Prototype Hardware and IoT Architecture

Component	Role in the prototype	Firmware interface
ESP32	Main controller, Wi-Fi connectivity and analog-data acquisition.	GPIO 34 (ACS712), GPIO 35 (ZMPT101B); Wi-Fi
ACS712	Measures AC current through the selected load path.	ACS712 library and automatic midpoint calibration
ZMPT101B	Measures AC RMS voltage through an isolated voltage-sensor module.	ZMPT101B library with configured sensitivity
16x2 I2C LCD	Displays local voltage, current, power and units.	I2C address 0x27
Blynk Cloud	Provides mobile dashboard presentation of live values.	Virtual pins V0-V3
EEPROM	Persists the cumulative energy value in local non-volatile memory.	EEPROM address 0; 512-byte allocation

Data path

AC load	ACS712 + ZMPT101B	ESP32	LCD / Serial	Wi-Fi / Blynk
---------	-------------------	-------	--------------	---------------

Safety requirement

The prototype involves mains-related sensing. Final work must use appropriate isolation, fuse protection, enclosed terminals, strain relief and supervision by a qualified person. The breadboard configuration should not be used as a final mains-connected product.

3. Evidence of Working Prototype

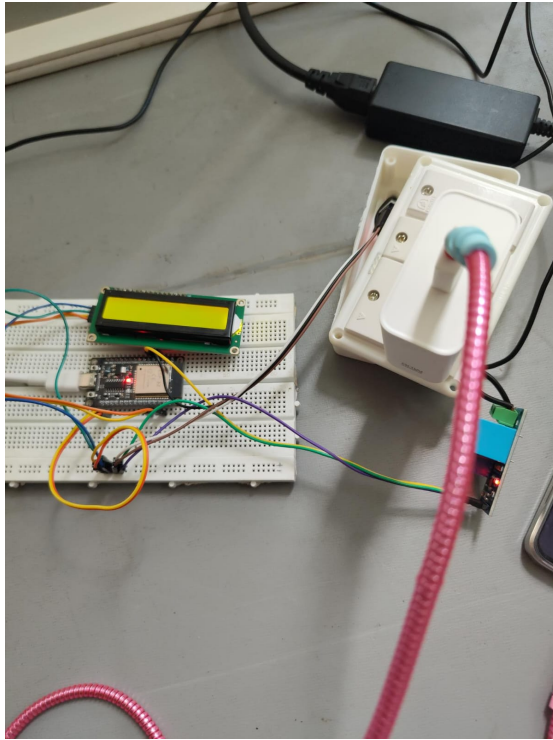


Figure 1. ESP32, LCD, sensor modules and connected load during prototype setup.

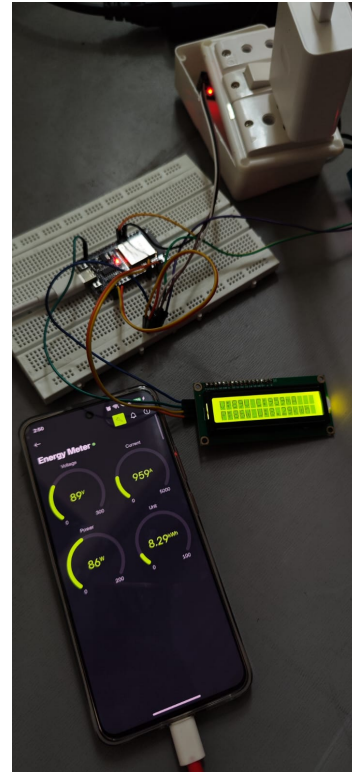


Figure 2. Live Blynk dashboard alongside the prototype.

Observed interface behaviour

The mobile dashboard displays Voltage, Current, Power and Unit values. The firmware updates the LCD and Blynk virtual pins after each measurement cycle. This confirms end-to-end sensing, local display and cloud dashboard integration at the prototype stage.

4. Preliminary Measurement Evidence

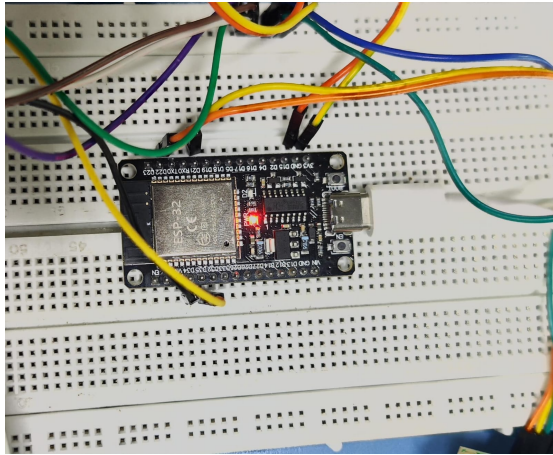


Figure 3. ESP32 controller mounted on the breadboard.

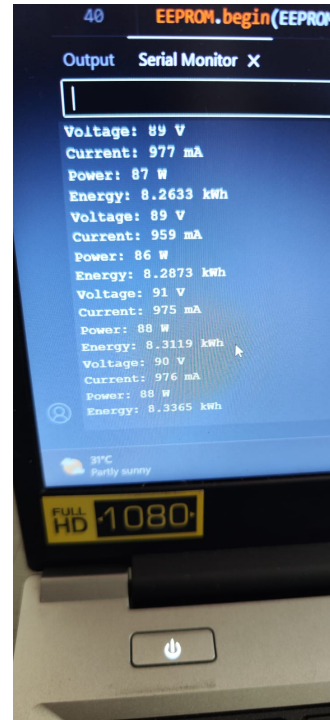


Figure 4. Serial-monitor output from the running firmware.

Readings visible in the serial monitor

Sample	Voltage	Current	Power	Displayed unit
1	89 V	977 mA	87 W	8.2633 kWh
2	89 V	959 mA	86 W	8.2873 kWh
3	91 V	975 mA	88 W	8.3119 kWh
4	90 V	976 mA	88 W	8.3365 kWh

Interpretation

These values are useful functional evidence that the sensor pipeline is active and that values are propagated to software interfaces. They are preliminary prototype readings, not yet calibrated measurements suitable for billing or model training.

5. Firmware Design and Validation Notes

The firmware initializes EEPROM, auto-calibrates the ACS712 midpoint, configures ZMPT101B sensitivity, connects the ESP32 to Blynk, reads voltage/current, calculates power, updates LCD/serial output, saves the unit value, and writes readings to Blynk V0-V3.

Representative firmware excerpt

```
// Credentials intentionally redacted from report
#define BLYNK_TEMPLATE_ID "[REDACTED]"
#define BLYNK_AUTH_TOKEN "[REDACTED]"

ACS712 ACS(34, 3.3, 4095, 125);
ZMPT101B voltageSensor(35, 50.0);

float voltage = voltageSensor.getRmsVoltage();
float watt = voltage * (mA / 1000.0);
Blynk.virtualWrite(V0, volt);
Blynk.virtualWrite(V1, current);
Blynk.virtualWrite(V2, power);
Blynk.virtualWrite(V3, unit);
```

Validation item	Status	Comment
Sensor-to-dashboard data flow	Demonstrated	Blynk dashboard and serial monitor show the same parameter classes.
EEPROM persistence	Implemented	Firmware reads/writes the unit value; restart persistence should be tested explicitly.
Voltage/current calibration	Required	Compare readings with a trusted meter across several loads.
Energy calculation	Requires correction	Energy must include elapsed time and correct Wh-to-kWh conversion before data logging.

Important technical correction before dataset collection

The present loop adds watt / 3600 on every 500 ms cycle. This does not use the actual time interval and labels the result as kWh. A correct approach is $\text{energy_kWh} += \text{power_W} * \text{elapsed_seconds} / 3,600,000$. Use `millis()` to calculate `elapsed_seconds`. Do not treat the current displayed unit total as calibrated energy data until this change and meter-based calibration are completed.

6. Dataset and Prediction Research Plan

Once the corrected firmware logs time-stamped readings, the project can produce a valid local dataset for analysis. The initial model should be evaluated against a simple baseline before claiming prediction capability.

Stage	Method	Output
Data capture	Log timestamp, voltage, current, power, incremental energy, device ID and optional tariff every 1-5 minutes.	CSV/database time series
Data quality	Remove gaps/outliers; validate daily kWh against a trusted reference meter.	Calibrated daily totals
Feature engineering	Daily lag values, weekday/weekend, previous 7-day mean, hourly-load profile; optional weather.	Model-ready dataset
Baseline	Predict tomorrow using today or same weekday last week.	Baseline MAE/RMSE
ML comparison	Train Linear Regression and Random Forest on past days; use a time-ordered train/test split.	Next-day kWh forecast

Public benchmark datasets reviewed

The UCI Appliances Energy Prediction dataset provides 19,735 10-minute observations with appliance-energy, indoor environment and weather features. The UCI ElectricityLoadDiagrams20112014 dataset provides 15-minute electricity values for 370 clients. These will be used only for exploratory modelling and pipeline testing; final project evaluation should use the project's calibrated ESP32 data.

Sources: UCI Appliances Energy Prediction, <https://archive-beta.ics.uci.edu/dataset/374/appliances%2Benergy%2Bprediction>; UCI ElectricityLoadDiagrams20112014, <https://archive.ics.uci.edu/dataset/321/electricityloadaddiagrams20112014>

7. Conclusion and Next Milestones

The current work has progressed beyond concept selection: an ESP32-based prototype has demonstrated electrical parameter acquisition, LCD output, serial output, EEPROM-backed cumulative value storage and Blynk dashboard communication. The supplied photographs and serial monitor provide documented evidence of this stage.

The next milestone is measurement validity

Before analytics or prediction is performed, the energy integration logic must be corrected and the voltage/current readings calibrated against a known reference meter. After this, structured time-stamped data can be collected over multiple weeks for genuine historical analysis and next-day forecasting.

Priority	Next action	Success criterion
1	Correct time-based kWh computation and verify EEPROM persistence.	Unit increment matches reference energy over a timed test.
2	Calibrate voltage/current sensor sensitivity using known loads and a multimeter/power meter.	Documented error percentage across test loads.
3	Add timestamped CSV/database logging while retaining Blynk updates.	At least several weeks of complete daily data.
4	Build dashboard analytics and evaluate next-day forecast against baseline.	MAE/RMSE and predicted-vs-actual chart.

Thank you